

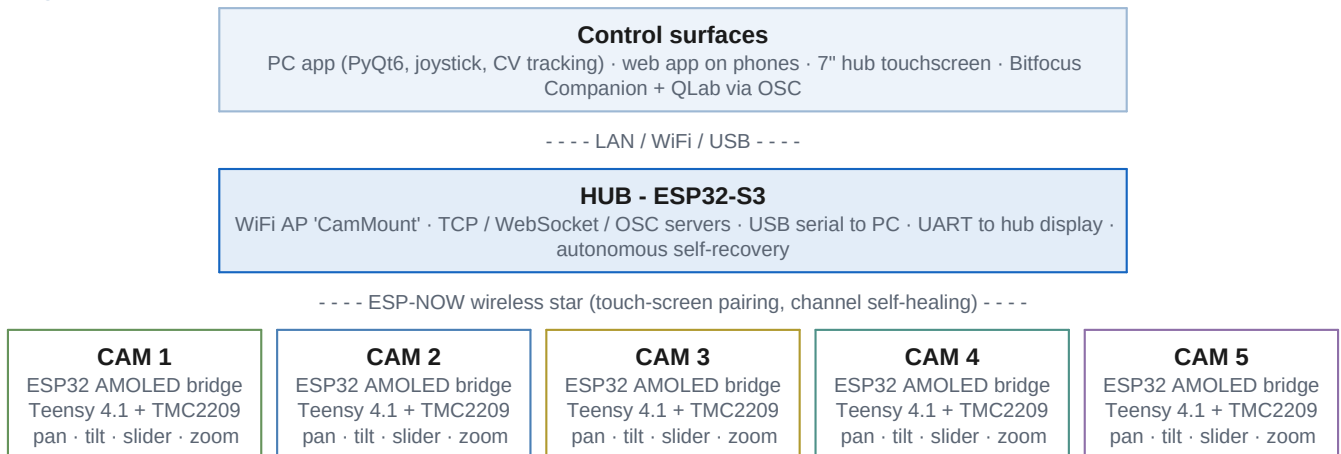
PTS

Pan / Tilt / Slider

A five-mount motorised camera system for live video production - wireless, self-healing, and controllable from everything in the booth.

PTS drives five motorised camera mounts over a dedicated wireless network. Each mount pairs a Teensy 4.1 motion controller (TMC2209 silent stepper drivers, StallGuard homing) with an ESP32 bridge and round touchscreen; a central hub links every mount to the operator's control surfaces. Positions, speed presets and subject calibrations live on the mounts themselves - controllers can come and go mid-show.

System architecture



Highlights

Feature	What it does
Look-at tracking	Camera keeps aiming at a calibrated 3D subject while the slider travels - subjects are calibrated in two taps and switchable mid-move from any controller.
Zero-config pairing	No MAC addresses or per-unit firmware: hold a mount's screen, pick CAM 1-5, tap your hub. The hub learns mounts on first contact; conflicts resolve with one tap on the hub display.
Self-healing	Autonomous recovery ladders on hub and mounts (reinit then restart), idle maintenance restarts, watchdogs at every layer, and a motion-side dead-man: a mount stops within 500 ms of losing its control stream.
Health telemetry	Every node reports heap, loop timing, link quality and reset cause every 10 s into the PC log - degradation is visible hours before it becomes a symptom.
Show-control ready	OSC servers on both the hub and the PC app put every function on Stream Deck buttons and QLab cue stacks (reference overleaf).

OSC control reference - Companion / QLab

Two identical OSC servers listen on **UDP port 9700** and accept the same address space, so Companion pages work unchanged against either target:

Target	Address	When to use
The hub itself	169.254.22.22	No PC needed - hub + display + mounts + Stream Deck is a complete rig. Reach it via a WiFi-to-LAN bridge joined to the CamMount AP, or any machine joined to CamMount directly.
PC app machine	that PC's IP	When the PC app is running anyway. Configurable via <code>osc_enabled</code> / <code>osc_port</code> in the app config.

Companion: add a **Generic: OSC** connection with the target IP and port 9700, then use its Send message actions as below. Integer arguments preferred; floats (QLab network cues) are accepted.

Command reference

Address	Arguments	Action
<code>/pts/estop</code>	-	E-STOP every mount
<code>/pts/cam/N/estop</code>	-	E-STOP mount N
<code>/pts/cam/N/goto</code>	slot 1-10	Recall a stored position (uses the mount's active speed presets)
<code>/pts/cam/N/store</code>	slot 1-10	Store the current position
<code>/pts/cam/N/clear</code>	slot 1-10	Clear a stored position
<code>/pts/cam/N/jog</code>	pan tilt slider zoom	Velocities -1000..1000; re-streamed at 20 Hz until stopped
<code>/pts/cam/N/jog/stop</code>	-	Stop jogging (put on the button's release action)
<code>/pts/cam/N/speed/pt</code>	1-4	Active pan/tilt speed preset
<code>/pts/cam/N/speed/sl</code>	1-4	Active slider speed preset
<code>/pts/cam/N/subject</code>	0-7	Select look-at subject; switches live during a move
<code>/pts/cam/N/lookat</code>	0 or 1	Look-at slider move to min (0) or max (1) with the selected subject

N = camera 1-5. Slots and speed presets are 1-based, subjects 0-based - matching every screen in the system.

Button recipes

Button	Setup
Recall shot 3, cam 2	Press: <code>/pts/cam/2/goto</code> with int 3
Hold-to-jog pan-left, cam 1	Press: <code>/pts/cam/1/jog</code> ints -400 0 0 0 Release: <code>/pts/cam/1/jog/stop</code>
Look-at pair, cam 3	Button A: <code>/pts/cam/3/subject</code> int 2 Button B: <code>/pts/cam/3/lookat</code> int 1 (press A mid-move to retarget live)
QLab cue	Network cue to port 9700: <code>/pts/cam/2/goto 4</code> - camera moves fire from the cue stack, in time with light and sound.
Show-stopper	A big red <code>/pts/estop</code> (no argument).

Safety

- OSC jogs are re-streamed at 20 Hz with a 15 s TTL, so a lost release message cannot run a mount away; prefer `goto` / `lookat` for long moves.
- The mounts' own 500 ms dead-man is the final backstop: if the hub or PC dies mid-jog, motion stops regardless of the controller.
- E-STOP over OSC clears held jog streams before broadcasting the stop.